

Chulin ZHAO

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Educational Background

Central South University (China) | University of Dundee (UK)

Sep. 2023- Jul. 2027(Expected)

Bachelor of Engineering in Computer Science (English-taught)

985 Project, Top-13 comprehensive in China; Computer Science: Top-11 nationwide, ESI Top 1% globally.

GPA: 89.44/100 Major Coursework GPA: 94/100

Core Courses: Algorithms and Artificial Intelligence; Object-Oriented Analysis and Design; Computer Networks and Data Communications; Database Systems; Information Security Technology; Agile Software Engineering;

Research Interests

Trustworthy AI in Healthcare, Uncertainty Quantification & Calibration, LLM Agents, Generative AI, Bioinformatics, Computational Imaging, Medical Imaging

Publication

- Zhao, C., Xu, Y., & Liu, S. (2026). Calibrated Closed-Form Uncertainty for Radiative Gaussian Splatting in Sparse-View CT. (<https://doi.org/10.48550/arXiv.2607.13682>)
- When Composition Doesn't Add Up: Humans Identifying Defects in AI-Generated Images (2026 IEEE International Workshop on Multimedia Signal Processing, Accepted)

Research Experiences

Closed-Form Uncertainty for Radiative Gaussian Splatting in Sparse-View CT

(Ongoing, in collaboration with Tencent, <https://arxiv.org/abs/2607.13682>)

Jun. 2026-Sep. 2026(Expected)

- Built a variational density posterior on R^2 -Gaussian. Exploiting strict linearity in X-ray rendering, closed-form predictive variance is computed exactly in a single forward pass, at substantially lower cost than sampling-based estimators.
- Conducted the first systematic calibration study for Gaussian-splatting CT (Spearman / AUSE / ECE with temperature scaling) on the official 15-scene benchmark across three view budgets: full-volume error ranking passes on 14/15 scenes.
- Proposed CGCA (coverage-gated + closed-form scoring) for active view acquisition: in anisotropic scenes, achieves view selection comparable to a 10-replica ensemble at roughly 1/10 the scoring cost.
- Core diagnosis: posterior variance is a constraint map, not an error map—high- σ regions reliably flag unconstrained tissue (trustworthy alarms), while low- σ regions do not certify accurate reconstruction (all-clears are not trustworthy).

LLM-Based Clinical Coding Framework (Ongoing)

(Research Assistant, Central South University)

April. 2026-Present

- Developed an uncertainty-aware, multi-label framework for mapping free-text EHRs to ICD codes through structured LLM reasoning.
- Modeled the ICD label space using co-occurrence-guided hierarchical partitioning to enable clinically coherent candidate pruning and efficient inference.
- Designed a bottom-up agentic pipeline that synthesizes raw clinical evidence and code relationships into executable routing and confirmation criteria.
- Formulated an LLM-driven inference mechanism with constrained decision actions, adaptive multi-branch reasoning, uncertainty-aware deferral, and traceable error attribution.

When Composition Doesn't Add Up: Humans Identifying Defects in AI-Generated Images

Jan. 2025-Nov. 2025

- Designed the CO-AID dataset targeting systematic defects in text-to-image models (Midjourney, Imagen, FLUX) under complex compositional prompts involving multiple entities, attributes, spatial relations, and interactions.
- Curated 651 reference images across four categories (people, hand, object, scene); refined compositional prompts through manual revision of ChatGPT-generated descriptions to emphasize multi-entity and relational complexity.

- Designed a two-tier defect annotation taxonomy (global-level and local-level); organized subjective experiment pipeline including training, pilot, and main study phases with 29 participants performing multi-label defect annotation.
- Fine-tuned TranSalNet on defect maps for local defect prediction; integrated predicted maps as spatial guidance into GPT-Image-1 for defect-guided image repair, achieving improved localization alignment with human annotations compared to direct GPT-Image-1 detection.

Honors & Awards

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| ● S Award of the 2025 MCM/ICM Mathematical Contest in Modeling | 2025 |
| ● Third-Place Academic Scholarship, Central South University | 2025 |